Part A: Probability & Conceptual Foundation

(Theory Questions and Answers)

1. What is Conditional Probability?

Conditional probability is the likelihood of an event occurring given that another event has already happened. In message classification, it represents the probability that a message belongs to a specific class (e.g., Spam) given the presence of certain observed features (e.g., specific keywords or URLs).

2. Explain Bayes' Theorem and its importance in classification problems.

Bayes' Theorem is a mathematical formula used to calculate conditional probability by updating our prior beliefs based on new evidence. It is crucial in classification because it allows models to compute the probability of a class label given the input features, forming the foundation for probabilistic models like Naive Bayes.

3. What assumptions does the Naive Bayes Classifier make?

The Naive Bayes Classifier operates on the "naive" assumption of conditional independence. It assumes that all input features are independent of one another given the target class, meaning the presence or value of one feature does not affect the probability of any other feature.

4. Explain the working principle of:

K-Nearest Neighbors (KNN): KNN classifies a new data point by measuring its distance to all training samples and identifying the 'K' closest neighbors. The point is then assigned to the class that is most common among those nearest neighbors (majority voting).

Support Vector Machine (SVM): SVM works by finding the optimal hyperplane that separates data points of different classes in a high-dimensional space. It aims to maximize the "margin," which is the distance between the hyperplane and the closest data points from each class (the support vectors).

5. Compare distance-based, probabilistic, and margin-based classifiers.

Distance-based classifiers (like KNN) rely on geometric proximity and similarity metrics to group data. Probabilistic classifiers (like Naive Bayes) use statistical likelihoods and Bayes' theorem to predict the most probable class. Margin-based classifiers (like SVM) focus on constructing optimal decision boundaries that maximize the separation space between different classes.

Source basis: Content provided in the user prompt. Citation markers were retained conceptually but not reproduced because the underlying citation references were not supplied.